

Lecture 3: Brain abscess and Poliomyelitis

A **brain abscess** is a pus-filled swelling in the brain. It usually occurs when bacteria or fungi enter the brain tissue after an infection or severe head injury. Although the risk of developing a brain abscess is extremely low, it is a life-threatening condition and should be diagnosed and treated as soon as possible.

Causes of a brain abscess

There are 3 main ways a brain abscess can develop. These are:

- An infection in another part of the skull – such as an ear infection, sinusitis or dental abscess, which can spread directly into the brain
- An infection in another part of the body – for example, the infection that causes pneumonia spreading into the brain via the blood
- Trauma, such as a severe head injury – those cracks open the skull, allowing bacteria or fungi to enter the brain

Symptoms of a brain abscess

The symptoms of a brain abscess may develop quickly or slowly but can include:

- **Headache** – which is often severe, located in a single section of the head and cannot be relieved with painkillers
- **Changes in mental state** – such as confusion or irritability
- problems with nerve function – such as muscle weakness, slurred speech or **paralysis** on one side of the body
- **High temperature**
- **Seizures (fits)**
- **Feeling sick**
- **Being sick**
- **Stiff neck**

- **Changes in vision** – such as blurring, greying of vision or [double vision](#) (due to the abscess putting pressure on the optic nerve)

Diagnosing a brain abscess

A brain abscess is regarded as a medical emergency. Swelling caused by the abscess can disrupt the blood and oxygen supply to the brain. There's also a risk of the abscess bursting (rupturing).

If left untreated, a brain abscess can cause permanent brain damage and could be fatal.

A brain abscess is usually treated using a combination of: Medicines – either antibiotics or antifungals

Surgery – either draining the pus through a hole in the skull (simple aspiration) or opening the skull and removing the abscess entirely (craniotomy)

Complications of a brain abscess

- Reoccurring abscess –this is more common in people with a weakened immune system or cyanotic heart disease
- Brain damage – mild to moderate brain damage often improves with time but severe brain damage is likely to be permanent; brain damage is more of a risk if diagnosis and treatment are delayed
- Epilepsy – where a person has repeated seizures (fits)
- Meningitis

POLIOMYELITIS

Poliomyelitis is an acute infectious disease that in its serious form affects the central nervous system. Ingested, spread by alimentary route: Commoner in areas of poor sanitation

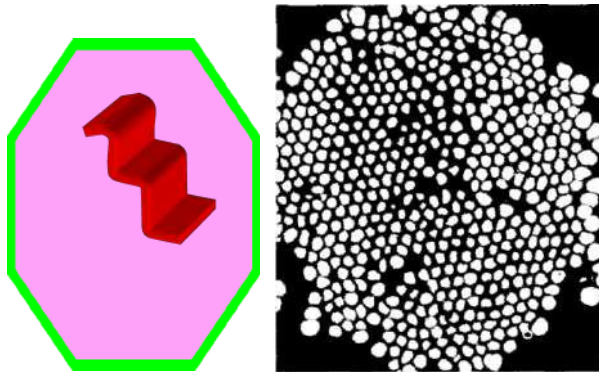
Infants protected by maternal antibodies

The destruction of motor neurons in the spinal cord results in flaccid paralysis (less than 0.1%). However, most poliovirus infections are subclinical.

Three antigenic types: Poliovirus I, II and III

During epidemic outbreaks, type I is most frequently isolated of cases while types II and III account for the remaining of cases.

Morphology "Picornavirus" from enteroviruses RNA viruses 20-30nm in diameter, positive sense, non-enveloped, 32 capsomeres, Capsid symmetry is of icosahedral



Optimal growth at 37°C

They are acid stable, it is not sensitive to the effect of ether and sodium desoxycholate

It is rapidly killed on boiling.

Pathogenesis

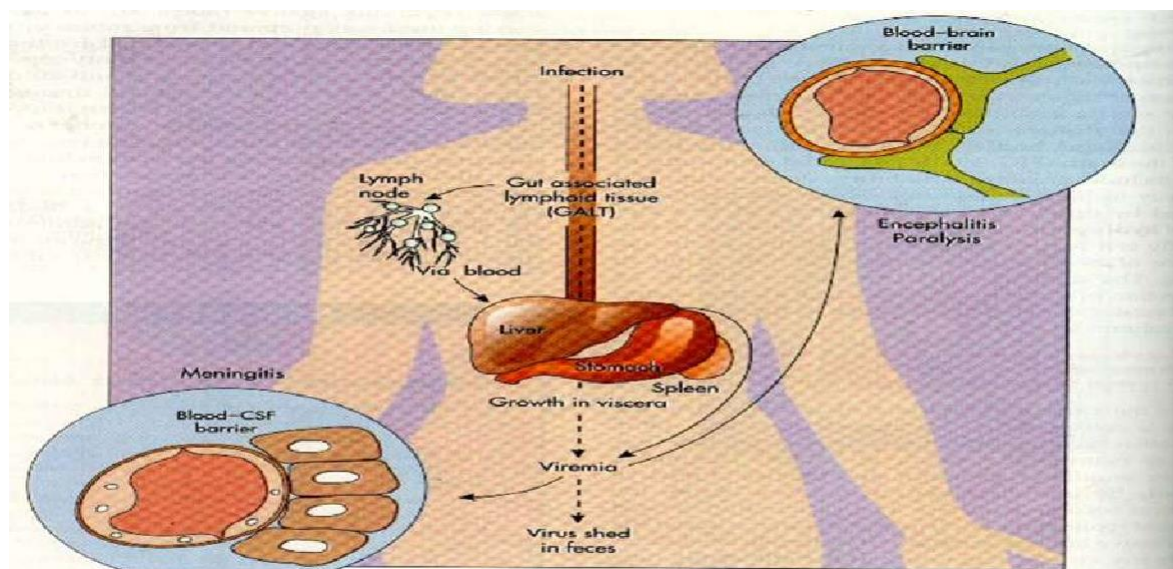


FIG. 19-4 Pathogenesis of poliomyelitis. Polioviruses enter the body through the gastrointestinal tract and spread elsewhere via blood and lymph. The virus can spread into the central nervous system (CNS) either via the blood-brain or blood-CSF barriers. Paralysis results from multiplication of polioviruses within the CNS.

Source of infection: Apparent and subclinical patients

Incubation: is usually 7-14 days, but it may range from 3 to 35 days.

Transmission

Fecal – oral route: poor hygiene, dirty diapers (especially in day-care settings)

Ingestion via contaminated food and water

Contact with infected hands

Inhalation of infectious aerosols

Clinical Findings

- Asymptomatic illness without symptoms 90% infected people
- Abortive poliomyelitis a minor illness 5 % of patients
- Aseptic meningitis – 1%-2% of patients with poliovirus infections,
- Paralytic poliomyelitis, the major illness 0.1% to 2% of persons with poliovirus

Only about 1% of infections are recognized clinically.

- Abortive Poliomyelitis. This is the commonest form of the disease. The patient has only the minor illness, characterized by fever, malaise, drowsiness, headache, nausea, vomiting, constipation, and sore throat in

various combinations. The patient recovers in a few days. The diagnosis of abortive poliomyelitis can be made only when the virus is isolated or antibody development is measured.

- Non paralytic Poliomyelitis (Aseptic Meningitis). In addition to the above symptoms and signs, the patient with the non-paralytic form presents as stiffness and pain in the back and neck with muscle spasm. The disease lasts 2-10 days, and recovery is rapid and complete. In a small percentage of cases, the disease advances to paralysis.
- Paralytic Poliomyelitis: The major illness usually follows the minor

illness. The predominating complaint is flaccid paralysis resulting from lower motor neuron damage. The maximal recovery usually occurs within 6 months, with residual paralysis lasting much longer. Bulbar poliomyelitis involves muscles of pharynx, vocal cords and respiration with death of 75% of patients

- Post-polio syndrome is a sequela of poliomyelitis occur 30-40- years with marked deterioration of the affected muscles due to loss of neurons in the initially affected nerves

Immunity Ig A and neutralizing antibody (Ig G, IgA, Ig M) persist for life span
Immunity is permanent to the type causing the infection.

Passive immunity is transferred from mother to offspring. The maternal antibodies gradually disappear during the first 6 months of life. Passively administered antibody lasts only 3-5 week

Lab Diagnosis

Definitive diagnosis is made by isolation of the virus from stool, CFS, oropharyngeal secretions

Cell culture involves fibroblastic MRC-5 cells

CPE is usually evident within 36 hours

Serotyping is based on neutralization of CPE

ELISA, IFA, neutralizing Test, CFT

Genomic detection by RT-PCR detection of viral RNA

Treatment

There is no specific treatment. Symptomatic relief and respiratory support involve reduction of pain and muscle spasm and maintenance of respiration and hydration. When the fever subsides, early mobilization and active exercise are begun. Analgesics, an orthopaedic regimen is set up from the first day that paralysis develops to prevent contractures and deformations, and exercise therapy

is carried out during the rehabilitation period. An apparatus for artificial respiration is employed when there are respiration disturbances.

Prevention

Both oral polio vaccine (OPV live, attenuated, Sabin, 1957) and inactivated poliovirus vaccine (IPV, Salk, 1954) are available

IPV is used for adult immunization and Immunocompromised patients Both killed and live virus vaccines induce antibodies and protect the central nervous system from subsequent invasion by wild virus

Advantages of OPV (Effectiveness-Lifelong immunity - Induction of secretory antibody response similar to that of natural infection - Possibility of attenuated virus circulating in community by spread to contacts (indirect immunization) (herd immunity)- Ease of administration - Lack of need for repeated boosters)

Disadvantages of OPV (Rarely of reversion of the attenuated virus to virulent one type 3 leads to vaccine derived polio virus VDPV- Infection of GIT with other enteroviruses limit viral replication and reduce protection - Unsafe for immunodeficient patients)

Advantages of IPV (Good stability in transport and during storage - Safe administration in immunodeficient patients - No risk of vaccine-related disease)

Disadvantages of IPV (Lack of induction of local (gut) immunity - Need for booster vaccine for lifelong immunity - Fact that injection is more painful than oral administration - Fact that higher community immunization levels are needed than with live vaccine)

Enhanced IPV is available with higher titer of antibody and induce some mucosal immunity but lower than OPV

Bivalent oral vaccine was produced by WHO to avoid high rate of reversion by type II.